Week 11 Manual Calculation: Example Questions and Model Answers

This file is a practice bank for the calculation section of Week 11. The questions focus on K-means, K-modes, distance calculation, centroid calculation, and reassignment.

Use this rule throughout:

```
K-means: numerical data -> mean centroid -> shortest distance wins
K-modes: categorical data -> mode center -> lowest dissimilarity wins
```

Question 1: Euclidean Distance Between Two Observations

Given:

Observation	x1	x2	x3	x4	x5
O1	10	2	-1	4	0
O2	12	4	-5	4	1

Calculate the Euclidean distance between O1 and O2.

Model Answer

Formula:

```
d(O1, O2) = sqrt((x1_1 - x1_2)^2 + (x2_1 - x2_2)^2 + ... + (x5_1 - x5_2)^2)
```

Substitute the values:

```
d(O1, O2)
= sqrt((10 - 12)^2 + (2 - 4)^2 + (-1 - -5)^2 + (4 - 4)^2 + (0 - 1)^2)
= sqrt((-2)^2 + (-2)^2 + (4)^2 + 0^2 + (-1)^2)
= sqrt(4 + 4 + 16 + 0 + 1)
= sqrt(25)
= 5
```

Final answer:

```
d(O1, O2) = 5
```

Question 2: Euclidean Distance With Negative Values

Given:

Observation	x1	x2	x3	x4	x5
O1	10	2	-1	4	0
O3	10	6	-6	4	0

Calculate the Euclidean distance between O1 and O3.

Model Answer

```
d(O1, O3)
= sqrt((10 - 10)^2 + (2 - 6)^2 + (-1 - -6)^2 + (4 - 4)^2 + (0 - 0)^2)
= sqrt(0^2 + (-4)^2 + 5^2 + 0^2 + 0^2)
= sqrt(0 + 16 + 25 + 0 + 0)
= sqrt(41)
= 6.403
```

Final answer:

```
d(O1, O3) = 6.403
```

Question 3: Manhattan Distance

Given:

Observation	x1	x2	x3
O1	10	2	-1
O2	12	4	-5

Calculate the Manhattan distance between O1 and O2.

Model Answer

Formula:

```
d(O1, O2) = |x1_1 - x1_2| + |x2_1 - x2_2| + |x3_1 - x3_2|
```

Substitute:

```
d(O1, O2)
= |10 - 12| + |2 - 4| + |-1 - -5|
= |-2| + |-2| + |4|
= 2 + 2 + 4
= 8
```

Final answer:

```
Manhattan distance = 8
```

Question 4: K-Means Centroid Calculation

Given cluster A contains:

Observation	x1	x2	x3	x4	x5
O1	10	2	-1	4	0
O4	9	2	-1	5	0
O7	8	4	-5	5	1

Calculate the centroid of cluster A.

Model Answer

For K-means, the centroid is the mean of each variable.

```
x1 = (10 + 9 + 8) / 3 = 27 / 3 = 9
x2 = (2 + 2 + 4) / 3 = 8 / 3 = 2.667
x3 = (-1 + -1 + -5) / 3 = -7 / 3 = -2.333
x4 = (4 + 5 + 5) / 3 = 14 / 3 = 4.667
x5 = (0 + 0 + 1) / 3 = 1 / 3 = 0.333
```

Final answer:

```
Centroid A = (9, 2.667, -2.333, 4.667, 0.333)
```

Question 5: Assign Observations to Nearest Centroid

Given the following observations:

Observation	x1	x2
O1	2	2

Observation	x1	x2
O2	3	4
O3	8	8
O4	9	6

Given centroids:

```
C1 = (2, 3)
C2 = (8, 7)
```

Using Euclidean distance, assign each observation to the nearest centroid.

Model Answer

Calculate distance to both centroids.

For O1:

```
d(O1, C1) = sqrt((2 - 2)^2 + (2 - 3)^2) = sqrt(1) = 1
d(O1, C2) = sqrt((2 - 8)^2 + (2 - 7)^2) = sqrt(36 + 25) = 7.810
```

For O2:

```
d(O2, C1) = sqrt((3 - 2)^2 + (4 - 3)^2) = sqrt(2) = 1.414
d(O2, C2) = sqrt((3 - 8)^2 + (4 - 7)^2) = sqrt(25 + 9) = 5.831
```

For O3:

```
d(O3, C1) = sqrt((8 - 2)^2 + (8 - 3)^2) = sqrt(36 + 25) = 7.810
d(O3, C2) = sqrt((8 - 8)^2 + (8 - 7)^2) = sqrt(1) = 1
```

For O4:

```
d(O4, C1) = sqrt((9 - 2)^2 + (6 - 3)^2) = sqrt(49 + 9) = 7.616
d(O4, C2) = sqrt((9 - 8)^2 + (6 - 7)^2) = sqrt(2) = 1.414
```

Assignment table:

Observation	d to C1	d to C2	Assigned cluster
O1	1.000	7.810	C1
O2	1.414	5.831	C1
O3	7.810	1.000	C2
O4	7.616	1.414	C2

Final answer:

```
C1 = O1, O2
C2 = O3, O4
```

Question 6: Recalculate K-Means Centroids After Assignment

Using the final clusters from Question 5:

```
C1 = O1, O2
C2 = O3, O4
```

with observations:

Observation	x1	x2
O1	2	2
O2	3	4
O3	8	8
O4	9	6

Recalculate the new centroids.

Model Answer

For C1:

```
C1 = O1, O2

x1 = (2 + 3) / 2 = 2.5
x2 = (2 + 4) / 2 = 3

New C1 = (2.5, 3)
```

For C2:

```
C2 = O3, O4

x1 = (8 + 9) / 2 = 8.5
x2 = (8 + 6) / 2 = 7

New C2 = (8.5, 7)
```

Final answer:

```
New C1 = (2.5, 3)
New C2 = (8.5, 7)
```

Question 7: One Full K-Means Iteration

Given:

i	x1	x2	x3	Initial group
1	10	2	-1	A
2	11	3	-1	B
3	18	5	-1	A
4	20	4	0	B
5	19	3	0	A
6	8	2	-1	B

First calculate the initial centroids. Then assign each observation to the nearest centroid using Euclidean distance.

Model Answer

Initial groups:

```
A = O1, O3, O5
B = O2, O4, O6
```

Centroid A:

```
x1 = (10 + 18 + 19) / 3 = 15.667
x2 = (2 + 5 + 3) / 3 = 3.333
x3 = (-1 + -1 + 0) / 3 = -0.667
```

```
CA = (15.667, 3.333, -0.667)
```

Centroid B:

```
x1 = (11 + 20 + 8) / 3 = 13
x2 = (3 + 4 + 2) / 3 = 3
x3 = (-1 + 0 + -1) / 3 = -0.667

CB = (13, 3, -0.667)
```

Distance and assignment:

i	d to CA	d to CB	New group
1	5.831	3.180	B
2	4.690	2.028	B
3	2.887	5.395	A
4	4.435	7.102	A
5	3.416	6.037	A
6	7.789	5.110	B

Final answer after one iteration:

```
A = O3, O4, O5
B = O1, O2, O6
```

Question 8: K-Means Stopping Rule

After one K-means reassignment, the clusters become:

```
A = O3, O4, O5
B = O1, O2, O6
```

The recalculated centroids are:

```
CA = (19, 4, -0.333)
CB = (9.667, 2.333, -1)
```

After calculating distances again, the new assignment is:

```
A = O3, O4, O5
B = O1, O2, O6
```

Should the K-means algorithm stop? Explain.

Model Answer

Yes, the algorithm should stop.

Reason:

```
The cluster assignments did not change after the new distance calculation.
```

In K-means, we repeat these two steps:

```
1. Assign observations to nearest centroid.
2. Recalculate centroids.
```

We stop when the assignments remain the same.

Final answer:

```
Stop, because the clusters are stable.
```

Question 9: K-Means Within-Cluster Sum of Squares

Given final clusters:

Observation	**x1**	**x2**	**Cluster**
O1	1	1	A
O2	2	1	A
O3	8	7	B
O4	9	8	B

The centroids are:

```
CA = (1.5, 1)
CB = (8.5, 7.5)
```

Calculate the within-cluster sum of squares, also called inertia.

Model Answer

For each observation, calculate squared distance to its own centroid. Do not take the square root for sum of squares.

For cluster A:

```
O1 to CA = (1 - 1.5)^2 + (1 - 1)^2 = 0.25 + 0 = 0.25
O2 to CA = (2 - 1.5)^2 + (1 - 1)^2 = 0.25 + 0 = 0.25
```

For cluster B:

```
O3 to CB = (8 - 8.5)^2 + (7 - 7.5)^2 = 0.25 + 0.25 = 0.50
O4 to CB = (9 - 8.5)^2 + (8 - 7.5)^2 = 0.25 + 0.25 = 0.50
```

Total:

```
inertia = 0.25 + 0.25 + 0.50 + 0.50 = 1.50
```

Final answer:

```
Within-cluster sum of squares = 1.5
```

Question 10: K-Modes Dissimilarity Score

Given:

Object	**x1**	**x2**	**x3**
O1	A	L	M
C1	A	L	C
C2	B	P	C

Calculate the dissimilarity score between O1 and C1, and between O1 and C2. Then assign O1 to a cluster.

Model Answer

For K-modes:

```
same = 0
different = 1
```

Compare O1 with C1:

Feature	Compare	Score
x1	A vs A	0
x2	L vs L	0
x3	M vs C	1

```
d(O1, C1) = 0 + 0 + 1 = 1
```

Compare O1 with C2:

Feature	Compare	Score
x1	A vs B	1
x2	L vs P	1
x3	M vs C	1

```
d(O1, C2) = 1 + 1 + 1 = 3
```

Since 1 is smaller than 3:

```
O1 is assigned to C1
```

Final answer:

```
d(O1, C1) = 1
d(O1, C2) = 3
O1 -> C1
```

Question 11: Full K-Modes Assignment Table

Given observations:

i	x1	x2	x3
1	A	L	M
2	B	P	C
3	A	L	C
4	B	L	C
5	A	P	M

Initial modes:

```
C1 = (A, L, C)
C2 = (B, P, C)
```

Calculate the dissimilarity to C1 and C2 for all observations.

Model Answer

Use:

```
same = 0
different = 1
```

i	Observation	d to C1	d to C2	Assigned cluster
1	(A, L, M)	1	3	C1
2	(B, P, C)	2	0	C2
3	(A, L, C)	0	2	C1
4	(B, L, C)	1	1	tie
5	(A, P, M)	2	2	tie

For O4:

```
O4 = (B, L, C)
C1 = (A, L, C) -> different, same, same -> 1 + 0 + 0 = 1
C2 = (B, P, C) -> same, different, same -> 0 + 1 + 0 = 1
```

For O5:

```
O5 = (A, P, M)
C1 = (A, L, C) -> same, different, different -> 0 + 1 + 1 = 2
```

```
C2 = (B, P, C) -> different, same, different -> 1 + 0 + 1 = 2
```

Final answer:

```
O1 -> C1
O2 -> C2
O3 -> C1
O4 -> tie
O5 -> tie
```

If the exam question gives a tie rule, apply that rule.

Question 12: K-Modes New Mode Calculation

Suppose a K-modes cluster contains:

Observation	x1	x2	x3
O1	A	L	M
O3	A	L	C
O5	A	P	M

Calculate the new mode for this cluster.

Model Answer

For K-modes, choose the most frequent category in each column.

For x1:

```
A, A, A -> A appears 3 times
mode x1 = A
```

For x2:

```
L, L, P -> L appears 2 times
mode x2 = L
```

For x3:

```
M, C, M -> M appears 2 times
mode x3 = M
```

Final answer:

```
New mode = (A, L, M)
```

Question 13: K-Modes Tie in Mode Calculation

Suppose a K-modes cluster contains:

Observation	x1	x2	x3
O1	A	L	M
O3	A	L	C
O4	B	L	C
O5	A	P	M

Calculate the new mode. Identify any tie.

Model Answer

For x1:

```
A, A, B, A
A appears 3 times, B appears 1 time
mode x1 = A
```

For x2:

```
L, L, L, P
L appears 3 times, P appears 1 time
mode x2 = L
```

For x3:

```
M, C, C, M
M appears 2 times, C appears 2 times
```

There is a tie for x3.

Final answer:

```
x1 = A
x2 = L
x3 = tie between M and C
```

So the new mode is:

```
(A, L, M) or (A, L, C), depending on the tie rule.
```

In an exam, write the tie clearly instead of hiding it.

Question 14: Identify the Correct Clustering Method

For each dataset, state whether K-means or K-modes is more suitable.

Dataset	Variables
A	age, income, spending score
B	color, brand, product type
C	height, weight, blood pressure
D	operating system, browser, country

Model Answer

K-means is used for numerical variables.

K-modes is used for categorical variables.

Dataset	Suitable method	Reason
A	K-means	age, income, and spending score are numerical
B	K-modes	color, brand, and product type are categorical
C	K-means	height, weight, and blood pressure are numerical
D	K-modes	operating system, browser, and country are categorical

Final answer:

```
A -> K-means
B -> K-modes
C -> K-means
D -> K-modes
```

Question 15: K-Prototypes Concept Question

A dataset contains these variables:

Variable	Type
Age	Numerical
Time Spent	Numerical
Operating System	Categorical
ISP	Categorical

Which clustering method is most suitable: K-means, K-modes, or K-prototypes? Explain.

Model Answer

K-means is mainly for numerical data.

K-modes is mainly for categorical data.

This dataset has both:

```
Numerical: Age, Time Spent
Categorical: Operating System, ISP
```

Therefore, the most suitable method is K-prototypes.

Final answer:

```
Use K-prototypes because the dataset contains both numerical and categorical
variables.
```

Final Exam Checklist

Before writing your final answer, check:

- Did I use mean for K-means?
- Did I use mode for K-modes?
- Did I square the differences for Euclidean distance?
- Did I use absolute values for Manhattan distance?
- Did I assign to the smallest distance or lowest dissimilarity?
- Did I recalculate the centroid or mode after reassignment?
- Did I clearly mention any tie?
- Did I stop only when the cluster assignment no longer changed?